

Instructions. (100 points) Sections 12.1–14.3. You may use a calculator. Show all work for credit.

- (10pts) 1. Let $A = (1, 3, -2)$, $B = (4, 1, 2)$, and $C = (2, 5, 0)$.

Find the scalar projection of $\overrightarrow{AC}$ onto $\overrightarrow{AB}$.

Solution:

$$\overrightarrow{AB} = \langle 3, -2, 4 \rangle, \quad \overrightarrow{AC} = \langle 1, 2, 2 \rangle$$

$$\overrightarrow{AC} \cdot \overrightarrow{AB} = 3 - 4 + 8 = 7$$

$$|\overrightarrow{AB}| = \sqrt{9 + 4 + 16} = \sqrt{29}$$

$$\text{comp}_{\overrightarrow{AB}} \overrightarrow{AC} = \frac{\overrightarrow{AC} \cdot \overrightarrow{AB}}{|\overrightarrow{AB}|} = \frac{7}{\sqrt{29}}$$

- (12pts) 2. Consider the two lines:

$$L_1 : \quad x = 2 + t, \quad y = 1 - t, \quad z = 3t$$

$$L_2 : \quad x = 4 + 2s, \quad y = -1 - 2s, \quad z = 6s$$

Are the lines **parallel**, **intersecting**, or **skew**? Justify your answer completely:

- If they are parallel, determine whether they are the **same line** or **two distinct lines**.
- If they intersect, find the point of intersection.

Solution:

Direction vectors: $\mathbf{d}_1 = \langle 1, -1, 3 \rangle$, $\mathbf{d}_2 = \langle 2, -2, 6 \rangle = 2\mathbf{d}_1$.

Since $\mathbf{d}_2$ is a scalar multiple of $\mathbf{d}_1$, the lines are **parallel**.

Now check if they are the same line. L_1 passes through $(2, 1, 0)$ (set $t = 0$). L_2 passes through $(4, -1, 0)$ (set $s = 0$).

Is $(4, -1, 0)$ on L_1 ? Solve $2 + t = 4 \Rightarrow t = 2$. Then $y = 1 - 2 = -1$ ($\checkmark$) and $z = 6 \neq 0$ ($\times$).

Since $(4, -1, 0) \notin L_1$, the lines are **parallel but distinct**.

(16^{pts}) **3.** Consider the surface $4x^2 - y^2 + 4z^2 = 16$.

(a) (8 pts) Find the family of traces in the most important direction and at least one trace in each of the other two coordinate directions. Use these to give a complete description of the surface.

Solution:

Divide by 16: $\frac{x^2}{4} - \frac{y^2}{16} + \frac{z^2}{4} = 1$

Trace $y = k$ (most important — the family): $\frac{x^2}{4} + \frac{z^2}{4} = 1 + \frac{k^2}{16}$ — **circles** of radius $2\sqrt{1 + k^2/16}$ for all values of k . The circles grow as $|k|$ increases. The smallest circle is at $y = 0$ with radius 2.

Trace $z = 0$ (xy -plane): $\frac{x^2}{4} - \frac{y^2}{16} = 1$ — **hyperbola** opening in the x -direction, vertices at $(\pm 2, 0, 0)$.

Trace $x = 0$ (yz -plane): $\frac{z^2}{4} - \frac{y^2}{16} = 1$ — **hyperbola** opening in the z -direction, vertices at $(0, 0, \pm 2)$.

This is a **hyperboloid of one sheet** with axis along the y -axis. The waist (smallest circle) is at $y = 0$ with radius 2.

(b) (8 pts) Carefully sketch the surface, labeling the intercepts and the axis of symmetry.

Solution:

Intercepts: $(\pm 2, 0, 0)$ and $(0, 0, \pm 2)$. No y -intercepts.

Axis of symmetry: the y -axis. The surface looks like a cooling tower with circular cross-sections, narrowest at $y = 0$ and widening as $|y|$ increases.

(16^{pts}) 4. Let $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 3t \rangle$.

(a) (4 pts) Show that the curve lies on a circular cylinder and give its equation.

Solution:

$x = 2 \cos t$ and $y = 2 \sin t$, so $x^2 + y^2 = 4 \cos^2 t + 4 \sin^2 t = 4$.

The curve lies on the cylinder $x^2 + y^2 = 4$ (radius 2, axis along z).

(b) (7 pts) Find the unit tangent vector $\mathbf{T}$ at $t = \pi/2$.

Solution:

$$\mathbf{r}'(t) = \langle -2 \sin t, 2 \cos t, 3 \rangle$$

$$|\mathbf{r}'(t)| = \sqrt{4 \sin^2 t + 4 \cos^2 t + 9} = \sqrt{4 + 9} = \sqrt{13}$$

$$\mathbf{r}'(\pi/2) = \langle -2, 0, 3 \rangle$$

$$\mathbf{T}(\pi/2) = \frac{\mathbf{r}'(\pi/2)}{|\mathbf{r}'(\pi/2)|} = \frac{1}{\sqrt{13}} \langle -2, 0, 3 \rangle = \left\langle \frac{-2}{\sqrt{13}}, 0, \frac{3}{\sqrt{13}} \right\rangle$$

(c) (5 pts) Find parametric equations for the tangent line to the curve at $t = \pi/2$.

Solution:

The point is $\mathbf{r}(\pi/2) = \langle 0, 2, 3\pi/2 \rangle$ and the direction is $\mathbf{r}'(\pi/2) = \langle -2, 0, 3 \rangle$.

$$x = -2s, \quad y = 2, \quad z = \frac{3\pi}{2} + 3s$$

- (12^{pts}) 5. Find a vector function $\mathbf{r}(t)$ that represents the curve of intersection of the cylinder $y^2 + z^2 = 4$ and the surface $x = yz$.

Solution:

The cylinder $y^2 + z^2 = 4$ is parameterized by $y = 2 \cos t$, $z = 2 \sin t$.

Then $x = yz = (2 \cos t)(2 \sin t) = 4 \cos t \sin t = 2 \sin 2t$.

$$\mathbf{r}(t) = \langle 2 \sin 2t, 2 \cos t, 2 \sin t \rangle, \quad 0 \leq t \leq 2\pi$$

(12^{pts}) 6. Consider the limit:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x y^4}{x^4 + y^4}$$

Does this limit **exist** or **not exist**? Provide careful evidence to support your conclusion. If it exists, prove it. If it does not exist, show why.

Solution:

The limit **exists and equals 0**.

Since $y^4 \leq x^4 + y^4$, we have $\frac{y^4}{x^4 + y^4} \leq 1$.

Therefore:

$$\left| \frac{x y^4}{x^4 + y^4} \right| = |x| \cdot \frac{y^4}{x^4 + y^4} \leq |x| \cdot 1 = |x|$$

As $(x, y) \rightarrow (0, 0)$, $|x| \rightarrow 0$, so by the Squeeze Theorem:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x y^4}{x^4 + y^4} = 0$$

- (12^{pts}) 7. For $g(r, s, t) = e^r \sin(st)$, find the mixed partial derivative g_{rst} .

Solution:

$$g_r = e^r \sin(st)$$

$$g_{rs} = e^r \cdot t \cos(st)$$

$$g_{rst} = e^r [\cos(st) + t(-\sin(st)) \cdot s] = e^r [\cos(st) - st \sin(st)]$$

(Product rule on $t \cos(st)$: derivative of t is 1, derivative of $\cos(st)$ w.r.t. t is $-s \sin(st)$.)

- (10^{pts}) 8. The equation $xz + yz^2 = 6$ defines z implicitly as a function of x and y near the point $(1, 1, 2)$.

Find $\frac{\partial z}{\partial x}$ at $(1, 1, 2)$.

Solution:

Differentiate both sides with respect to x (treating y as constant, $z = z(x, y)$):

$$z + x \frac{\partial z}{\partial x} + 2yz \frac{\partial z}{\partial x} = 0$$

$$\frac{\partial z}{\partial x}(x + 2yz) = -z \Rightarrow \frac{\partial z}{\partial x} = \frac{-z}{x + 2yz}$$

$$\text{At } (1, 1, 2): \frac{\partial z}{\partial x} = \frac{-2}{1 + 2(1)(2)} = \frac{-2}{5}$$